

Wave 1 Statistical Analyses

Marta

Hierarchical Bayesian Models Wave 1

This document runs the models used to analyse the main outcomes for Wave 1. The structure of the document follows the Analysis_Plans.txt file. The variables in the datasets are explained in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).
The explainer variables are: * Whether the participant was at a university that was affiliated with the UKRN
* Whether the participant was in their first, second, placement or third year
* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean  sd   2.5%   25%   50%   75%  97.5% n_eff
## mu          0.10    0.01 0.44  -0.78  -0.19   0.10   0.39   0.96 2881
## sigma       1.12    0.00 0.25   0.71   0.94   1.09   1.27   1.67 10124
## alpha[1]    -0.80    0.01 0.65  -2.10  -1.21  -0.79  -0.37   0.46 2779
## alpha[2]     0.91    0.01 0.78  -0.60   0.39   0.90   1.41   2.47 3900
## alpha[3]     0.27    0.01 0.70  -1.12  -0.20   0.27   0.73   1.63 3228
## alpha[4]     0.27    0.01 0.73  -1.18  -0.22   0.27   0.75   1.73 3482
## alpha[5]    -1.38    0.01 0.76  -2.92  -1.87  -1.35  -0.87   0.08 3738
## alpha[6]     0.87    0.01 0.82  -0.69   0.32   0.85   1.39   2.55 4481
## alpha[7]    -1.21    0.01 0.77  -2.77  -1.71  -1.19  -0.69   0.25 4104
## alpha[8]     1.88    0.01 0.83   0.35   1.31   1.85   2.41   3.63 4515
## alpha[9]    -0.09    0.01 0.58  -1.24  -0.49  -0.09   0.30   1.05 7007
## alpha[10]    2.02    0.01 0.87   0.51   1.41   1.96   2.55   3.93 8453
## alpha[11]    0.55    0.01 0.71  -0.81   0.07   0.53   1.01   1.98 7387
## alpha[12]   -1.44    0.01 0.76  -3.03  -1.93  -1.40  -0.93  -0.07 9282
## alpha[13]    0.38    0.01 0.53  -0.64   0.03   0.39   0.73   1.43 6847
## alpha[14]    0.00    0.01 0.52  -1.04  -0.36   0.00   0.35   1.02 6659
## alpha[15]   -1.26    0.01 0.69  -2.68  -1.70  -1.24  -0.79   0.02 8109
## alpha[16]    0.97    0.01 0.44   0.13   0.67   0.96   1.26   1.86 5579
## alpha[17]   -0.17    0.01 0.48  -1.12  -0.49  -0.17   0.16   0.78 7038
## alpha[18]   -1.18    0.01 0.47  -2.13  -1.49  -1.17  -0.86  -0.31 7195
## alpha[19]    0.27    0.01 0.56  -0.82  -0.11   0.27   0.65   1.40 5742
## alpha[20]    0.54    0.01 0.40  -0.23   0.26   0.53   0.80   1.32 4416
## alpha[21]    0.38    0.01 0.41  -0.42   0.10   0.37   0.65   1.20 5363
## alpha[22]    0.42    0.01 0.41  -0.37   0.14   0.41   0.69   1.23 4591
## alpha[23]    0.11    0.01 0.66  -1.17  -0.34   0.10   0.54   1.41 8603
```

```

## beta1      -0.38    0.01 0.70   -1.74   -0.85   -0.39    0.07    1.00  2953
## beta2       0.07    0.00 0.15   -0.22   -0.03    0.07    0.17    0.35  2980
## beta3       0.80    0.00 0.24    0.35    0.64    0.80    0.96    1.28  5470
## lp__      -477.27    0.05 4.14 -486.34 -479.83 -476.92 -474.30 -470.35  5985
##           Rhat
## mu          1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1
## alpha[10]    1
## alpha[11]    1
## alpha[12]    1
## alpha[13]    1
## alpha[14]    1
## alpha[15]    1
## alpha[16]    1
## alpha[17]    1
## alpha[18]    1
## alpha[19]    1
## alpha[20]    1
## alpha[21]    1
## alpha[22]    1
## alpha[23]    1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 19:47:21 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 803

UKRN N = 421

Non-UKRN N = 382

Year 1 N = 326

Year 2 N = 261

Placement Year N = 57

Year 4 N = 159

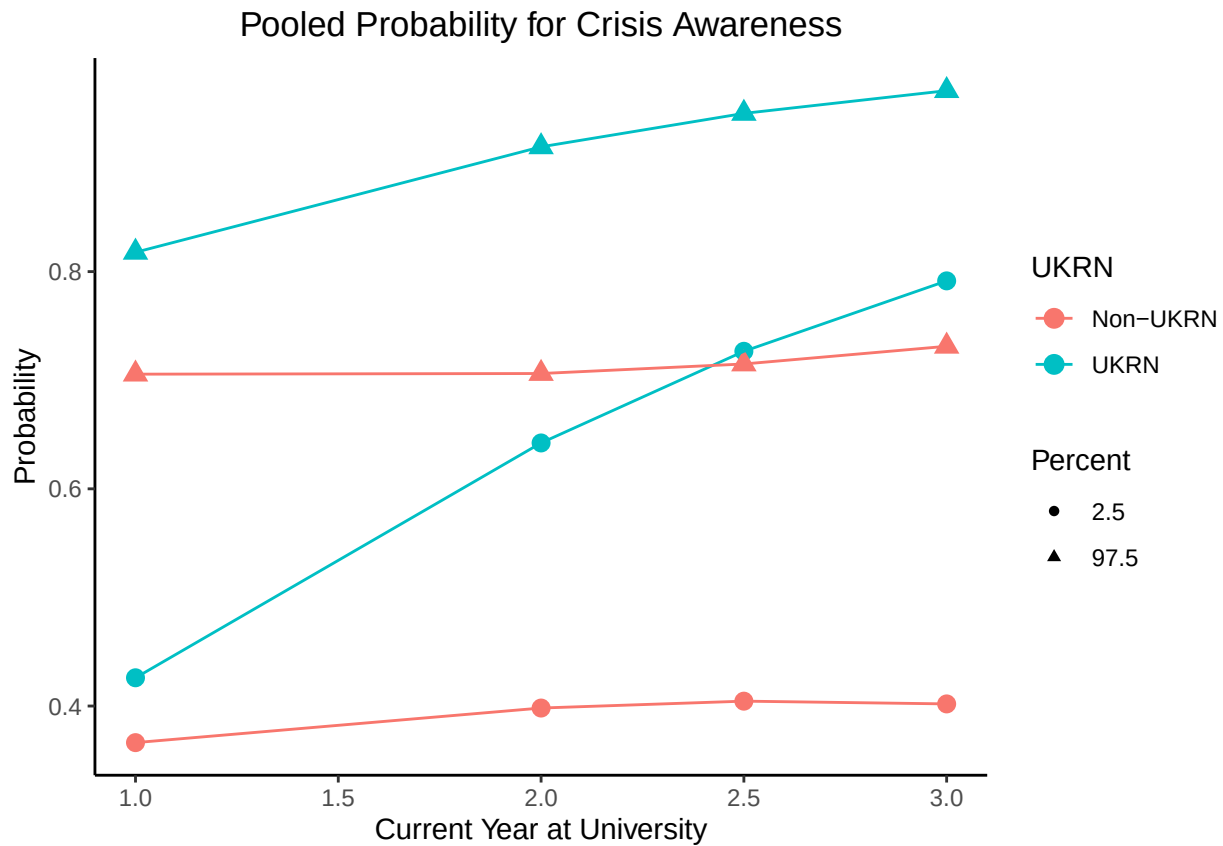
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.3464113

97.5% = 1.2778323

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the reproducibility crisis in the later years of education. For students who attend a non-UKRN affiliated

university, there is no difference.



Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement of third year

* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           -2.07    0.01 0.49   -3.07   -2.39   -2.06   -1.74   -1.14  2105
## sigma         0.77    0.00 0.21    0.43    0.62    0.75    0.89    1.24  5219
## alpha[1]     -2.64    0.01 0.60   -3.84   -3.04   -2.63   -2.24   -1.47  2391
## alpha[2]     -1.66    0.01 0.66   -2.96   -2.10   -1.65   -1.22   -0.35  2773
## alpha[3]     -2.22    0.01 0.64   -3.49   -2.64   -2.22   -1.80   -0.98  2675
## alpha[4]     -1.60    0.01 0.66   -2.90   -2.03   -1.60   -1.17   -0.32  2870
## alpha[5]     -2.92    0.01 0.78   -4.57   -3.41   -2.89   -2.38   -1.48  3641
## alpha[6]     -1.40    0.01 0.68   -2.73   -1.85   -1.41   -0.96   -0.04  3029
## alpha[7]     -3.05    0.01 0.75   -4.61   -3.54   -3.02   -2.52   -1.66  3363
## alpha[8]     -1.08    0.01 0.63   -2.31   -1.50   -1.08   -0.66    0.15  2535
## alpha[9]     -1.59    0.01 0.66   -2.89   -2.02   -1.59   -1.14   -0.30  3822
## alpha[10]    -1.54    0.01 0.67   -2.87   -1.98   -1.53   -1.09   -0.24  4049
```

```

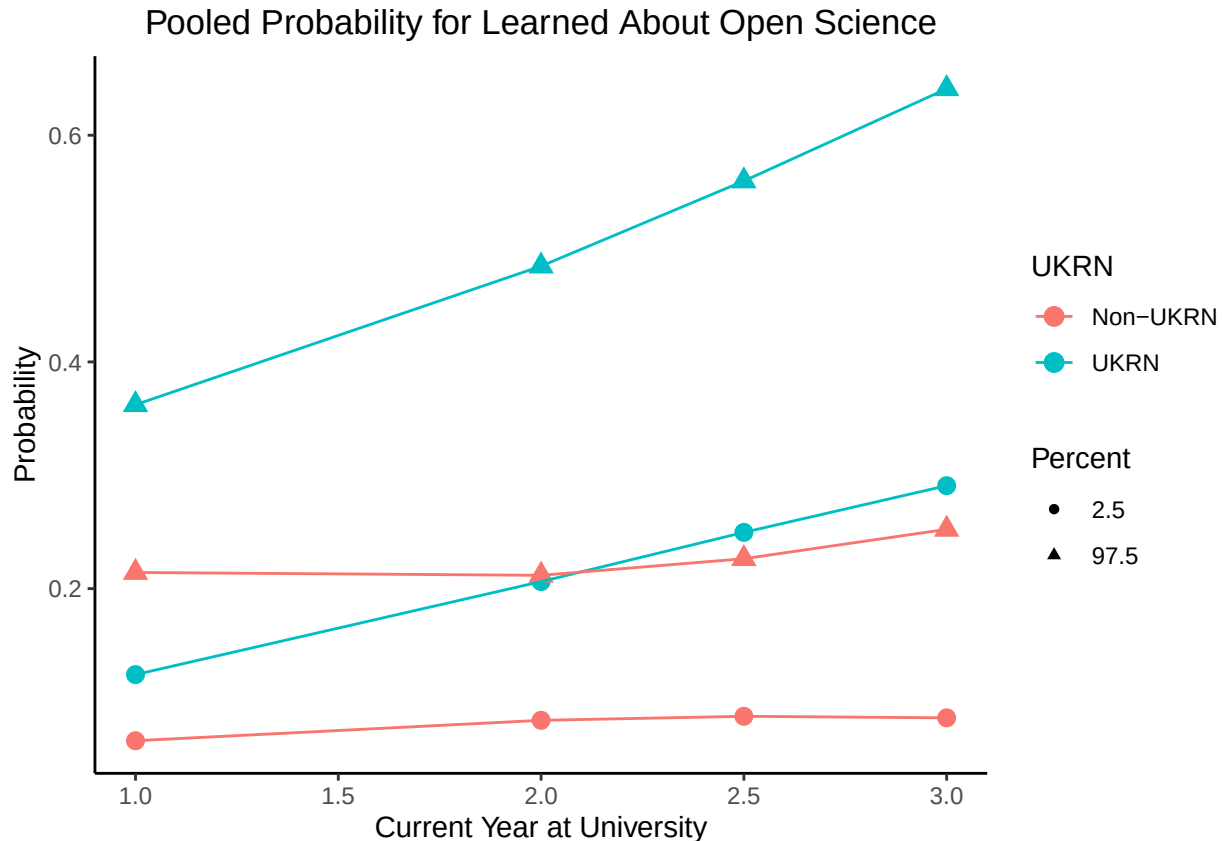
## alpha[11] -2.67 0.01 0.85 -4.45 -3.20 -2.62 -2.10 -1.14 4422
## alpha[12] -2.64 0.01 0.82 -4.37 -3.15 -2.59 -2.08 -1.18 4812
## alpha[13] -2.00 0.01 0.64 -3.29 -2.42 -1.99 -1.56 -0.79 4079
## alpha[14] -2.27 0.01 0.69 -3.70 -2.72 -2.25 -1.79 -0.98 4274
## alpha[15] -2.03 0.01 0.71 -3.48 -2.50 -2.02 -1.55 -0.66 4288
## alpha[16] -2.24 0.01 0.57 -3.41 -2.60 -2.22 -1.85 -1.19 3516
## alpha[17] -1.65 0.01 0.58 -2.81 -2.04 -1.64 -1.26 -0.53 3935
## alpha[18] -3.10 0.01 0.76 -4.74 -3.57 -3.05 -2.58 -1.75 4769
## alpha[19] -1.80 0.01 0.67 -3.15 -2.25 -1.79 -1.35 -0.51 3728
## alpha[20] -1.76 0.01 0.51 -2.78 -2.10 -1.75 -1.41 -0.77 2908
## alpha[21] -1.51 0.01 0.51 -2.51 -1.84 -1.50 -1.17 -0.54 3352
## alpha[22] -2.51 0.01 0.57 -3.67 -2.89 -2.49 -2.12 -1.43 3359
## alpha[23] -1.81 0.01 0.74 -3.29 -2.30 -1.79 -1.32 -0.38 4517
## beta1 0.28 0.01 0.66 -1.00 -0.16 0.28 0.72 1.57 2674
## beta2 0.12 0.00 0.20 -0.26 -0.01 0.12 0.26 0.51 2240
## beta3 0.43 0.00 0.25 -0.07 0.25 0.43 0.60 0.93 3307
## lp__ -384.83 0.07 4.82 -395.16 -387.83 -384.57 -381.47 -376.30 4333
## Rhat
## mu 1
## sigma 1
## alpha[1] 1
## alpha[2] 1
## alpha[3] 1
## alpha[4] 1
## alpha[5] 1
## alpha[6] 1
## alpha[7] 1
## alpha[8] 1
## alpha[9] 1
## alpha[10] 1
## alpha[11] 1
## alpha[12] 1
## alpha[13] 1
## alpha[14] 1
## alpha[15] 1
## alpha[16] 1
## alpha[17] 1
## alpha[18] 1
## alpha[19] 1
## alpha[20] 1
## alpha[21] 1
## alpha[22] 1
## alpha[23] 1
## beta1 1
## beta2 1
## beta3 1
## lp__ 1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 19:49:56 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 798
 UKRN N = 420
 Non-UKRN N = 378
 Year 1 N = 324
 Year 2 N = 258
 Placement Year N = 57
 Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. The results are non-significant.



Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement of third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5%  n_eff
## mu           -1.36     0.01  0.38   -2.08   -1.61   -1.36   -1.11   -0.61  1152
## sigma         0.38     0.00  0.15    0.14    0.27    0.36    0.46    0.71  1984
## alpha[1]     -1.68     0.01  0.44   -2.56   -1.98   -1.68   -1.39   -0.81  1380
## alpha[2]     -1.47     0.01  0.49   -2.43   -1.79   -1.47   -1.14   -0.53  1687
```

```

## alpha[3]      -1.45      0.01 0.46   -2.37   -1.76   -1.45   -1.14   -0.54  1574
## alpha[4]      -1.15      0.01 0.48   -2.07   -1.48   -1.16   -0.84   -0.20  1711
## alpha[5]      -1.57      0.01 0.51   -2.63   -1.90   -1.56   -1.23   -0.61  1687
## alpha[6]      -1.07      0.01 0.50   -2.00   -1.41   -1.09   -0.75   -0.07  1807
## alpha[7]      -1.50      0.01 0.49   -2.49   -1.82   -1.49   -1.17   -0.54  1704
## alpha[8]      -0.96      0.01 0.47   -1.85   -1.28   -0.96   -0.65   -0.02  1538
## alpha[9]      -1.28      0.01 0.48   -2.22   -1.60   -1.28   -0.96   -0.32  2006
## alpha[10]     -1.36      0.01 0.50   -2.35   -1.70   -1.36   -1.04   -0.38  1884
## alpha[11]     -1.40      0.01 0.51   -2.44   -1.74   -1.39   -1.06   -0.40  1975
## alpha[12]     -1.64      0.01 0.54   -2.78   -1.97   -1.61   -1.28   -0.66  1958
## alpha[13]     -1.47      0.01 0.47   -2.43   -1.78   -1.46   -1.15   -0.57  1826
## alpha[14]     -1.47      0.01 0.48   -2.43   -1.79   -1.46   -1.14   -0.54  1864
## alpha[15]     -0.91      0.01 0.53   -1.88   -1.28   -0.93   -0.57    0.19  1901
## alpha[16]     -1.53      0.01 0.44   -2.40   -1.82   -1.53   -1.24   -0.68  1697
## alpha[17]     -1.43      0.01 0.46   -2.34   -1.73   -1.43   -1.12   -0.52  1770
## alpha[18]     -1.42      0.01 0.43   -2.28   -1.71   -1.42   -1.12   -0.60  1834
## alpha[19]     -1.58      0.01 0.51   -2.63   -1.90   -1.56   -1.24   -0.63  1819
## alpha[20]     -1.03      0.01 0.42   -1.85   -1.32   -1.03   -0.75   -0.22  1551
## alpha[21]     -1.17      0.01 0.42   -1.98   -1.44   -1.17   -0.88   -0.36  1735
## alpha[22]     -1.32      0.01 0.42   -2.15   -1.60   -1.32   -1.04   -0.51  1537
## alpha[23]     -1.34      0.01 0.51   -2.33   -1.68   -1.34   -1.00   -0.34  1952
## beta1         -0.04      0.01 0.51   -1.04   -0.38   -0.04    0.31    0.97  1668
## beta2          0.06      0.00 0.16   -0.26   -0.04    0.07    0.18    0.38  1282
## beta3          0.28      0.01 0.22   -0.16    0.13    0.28    0.43    0.71  1966
## lp__         -438.94      0.19 7.33 -452.55 -443.85 -439.29 -434.39 -423.17  1502
##
## Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## alpha[20]     1
## alpha[21]     1
## alpha[22]     1
## alpha[23]     1
## beta1         1
## beta2         1
## beta3         1

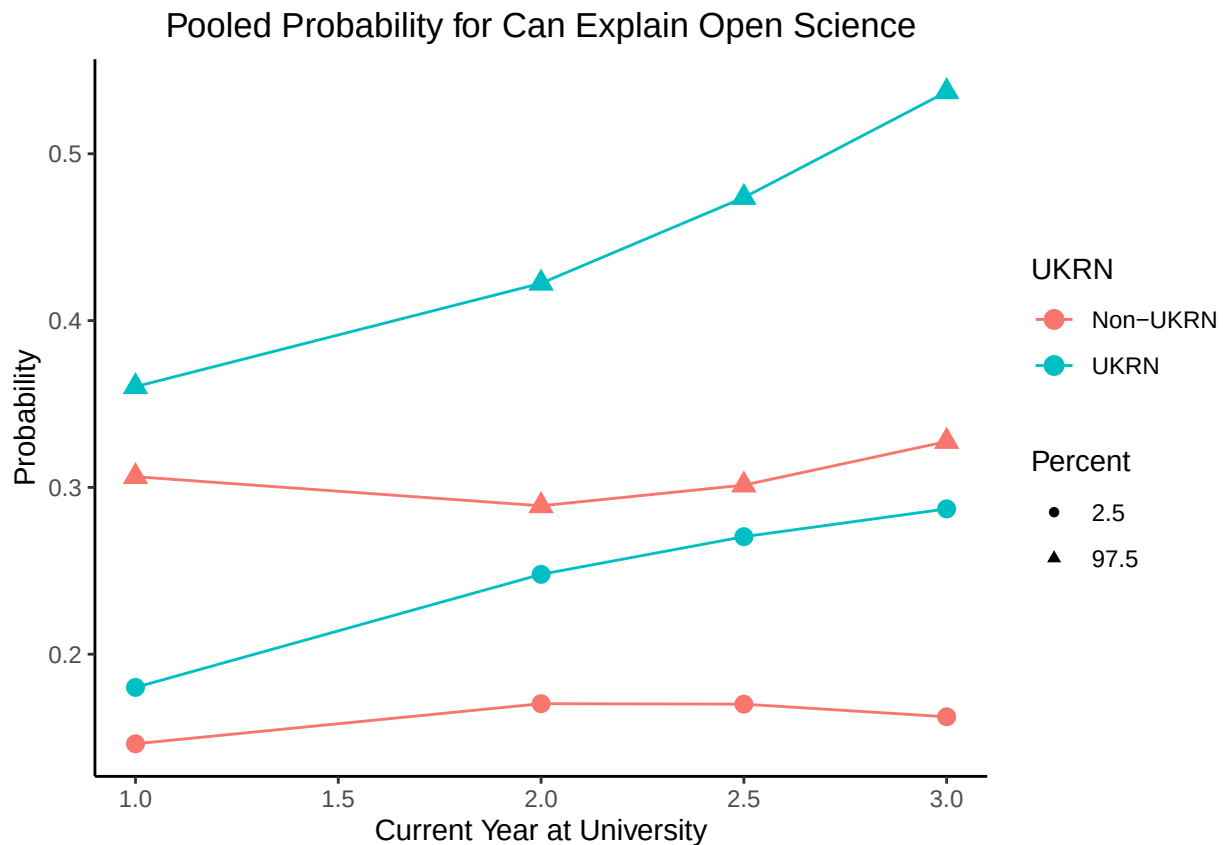
```

```
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan  6 19:52:06 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 798
 UKRN N = 420
 Non-UKRN N = 378
 Year 1 N = 324
 Year 2 N = 258
 Placement Year N = 57
 Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. No significant results.



Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no).

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement of third year
- * Interaction between the two variables above

```

## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##      mean se_mean  sd  2.5%  25%  50%  75%  97.5% n_eff
## mu      3.31    0.02 0.66   2.06  2.85  3.27  3.74   4.65 1499
## sigma    0.85    0.00 0.27   0.38  0.67  0.83  1.01   1.45 2947
## alpha[1] 3.00    0.02 0.80   1.46  2.47  2.98  3.52   4.61 1837
## alpha[2] 2.61    0.02 0.87   0.92  2.04  2.61  3.18   4.32 2125
## alpha[3] 3.49    0.02 0.87   1.85  2.90  3.46  4.05   5.26 2088
## alpha[4] 3.75    0.02 0.94   2.03  3.09  3.70  4.36   5.70 2349
## alpha[5] 2.54    0.02 0.89   0.80  1.96  2.52  3.12   4.29 2176
## alpha[6] 3.98    0.02 1.02   2.14  3.29  3.92  4.61   6.17 2746
## alpha[7] 3.39    0.02 0.95   1.58  2.73  3.35  4.01   5.34 2656
## alpha[8] 3.69    0.02 0.90   2.02  3.07  3.66  4.27   5.52 2183
## alpha[9] 3.40    0.02 0.89   1.73  2.79  3.37  3.98   5.24 2733
## alpha[10] 3.38    0.02 0.91   1.68  2.76  3.35  3.97   5.27 2767
## alpha[11] 3.33    0.02 0.94   1.57  2.69  3.29  3.92   5.28 2638
## alpha[12] 3.73    0.02 1.00   1.93  3.05  3.67  4.35   5.89 3097
## alpha[13] 2.05    0.02 0.78   0.59  1.51  2.03  2.55   3.64 2275
## alpha[14] 3.52    0.02 0.89   1.89  2.90  3.47  4.09   5.39 2600
## alpha[15] 3.34    0.02 0.91   1.64  2.72  3.30  3.92   5.24 2732
## alpha[16] 3.91    0.02 0.84   2.41  3.32  3.86  4.44   5.71 2631
## alpha[17] 3.51    0.02 0.85   1.97  2.93  3.48  4.06   5.28 2771
## alpha[18] 3.07    0.01 0.74   1.72  2.57  3.04  3.56   4.62 2531
## alpha[19] 2.62    0.02 0.83   1.05  2.06  2.60  3.15   4.29 2270
## alpha[20] 4.22    0.02 0.84   2.69  3.63  4.18  4.76   6.00 2398
## alpha[21] 4.24    0.02 0.94   2.61  3.58  4.17  4.82   6.29 2605
## alpha[22] 2.56    0.02 0.68   1.28  2.09  2.53  3.00   3.96 1957
## alpha[23] 2.72    0.02 0.91   0.97  2.11  2.71  3.32   4.55 2742
## beta1     -1.66    0.02 0.89  -3.43  -2.24  -1.66  -1.07   0.06 2053
## beta2     -0.36    0.01 0.26  -0.89  -0.53  -0.35  -0.19   0.14 1634
## beta3      0.88    0.01 0.38   0.14   0.62   0.87   1.13   1.65 2853
## lp__     -240.44    0.12 5.82 -252.02 -244.19 -240.43 -236.78 -228.65 2411
##      Rhat
## mu      1
## sigma    1
## alpha[1] 1
## alpha[2] 1
## alpha[3] 1
## alpha[4] 1
## alpha[5] 1
## alpha[6] 1
## alpha[7] 1
## alpha[8] 1
## alpha[9] 1
## alpha[10] 1
## alpha[11] 1
## alpha[12] 1
## alpha[13] 1
## alpha[14] 1
## alpha[15] 1
## alpha[16] 1
## alpha[17] 1

```

```

## alpha[18]      1
## alpha[19]      1
## alpha[20]      1
## alpha[21]      1
## alpha[22]      1
## alpha[23]      1
## beta1          1
## beta2          1
## beta3          1
## lp__           1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 19:54:28 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 798
UKRN N = 420
Non-UKRN N = 378
Year 1 N = 324
Year 2 N = 258
Placement Year N = 57
Year 4 N = 159

The output shows that beta credibility intervals do not cross 0 for the UKRN status and the interaction term.

UKRN Status

2.5% = -3.4268701

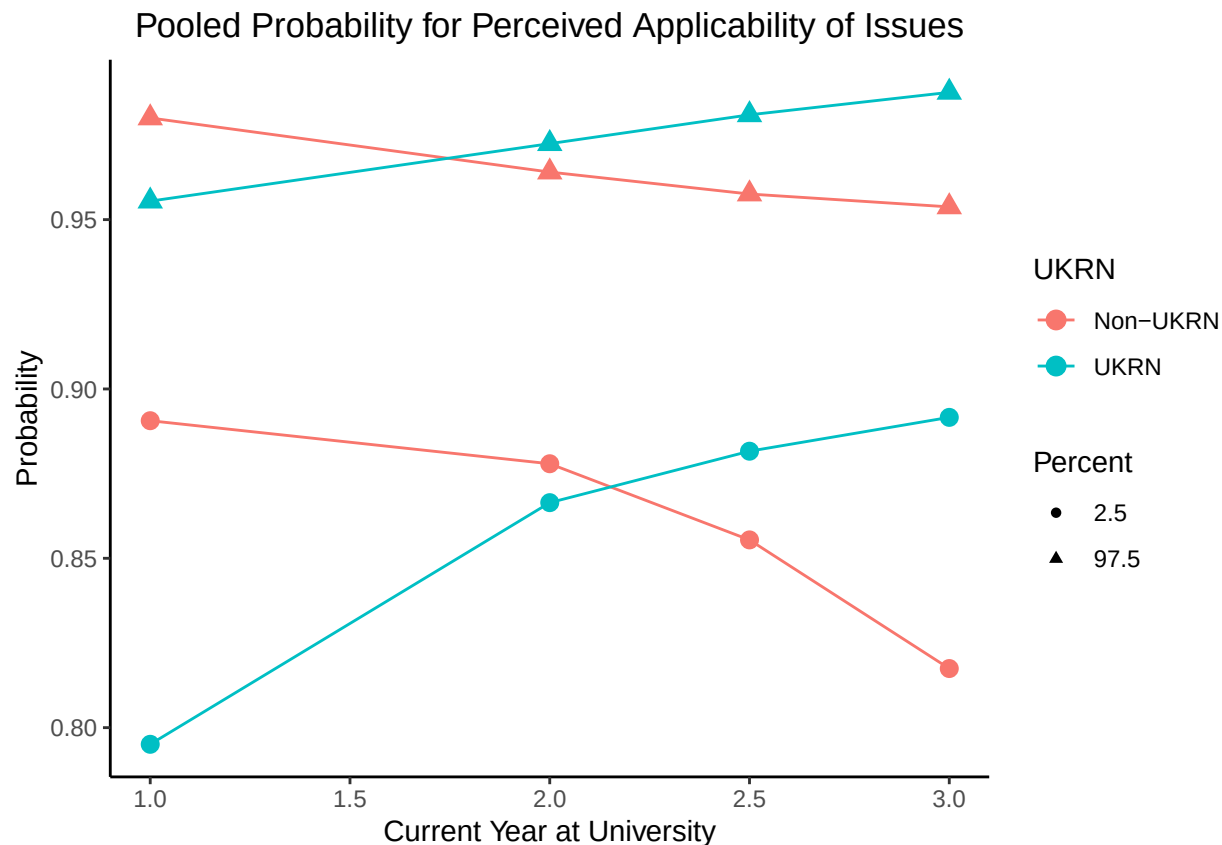
97.5% = 0.0637949 Interaction term:

2.5% = 0.1365337

97.5% = 1.6493499

The interpretation for the UKRN term, is that students who were in non-UKRN universities were more likely to think that the issues covered in the questionnaire were relevant to them.

The interpretation for the interaction term is that students who attend a UKRN affiliated university are more likely to think that the issues around open and reproducible research are applicable to their work in the later years of education.



Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8. The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

Inference for Stan model: anon_model.

4 chains, each with iter=10000; warmup=5000; thin=1;

post-warmup draws per chain=5000, total post-warmup draws=20000.

##

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff
## mu	1.54	0.00	0.08	1.38	1.48	1.54	1.59	1.70	1771
## sigma	0.12	0.00	0.04	0.05	0.09	0.11	0.14	0.20	3103
## alpha[1]	1.51	0.00	0.10	1.31	1.44	1.51	1.58	1.72	2182
## alpha[2]	1.54	0.00	0.12	1.30	1.46	1.54	1.62	1.78	2790
## alpha[3]	1.52	0.00	0.11	1.29	1.44	1.52	1.59	1.74	2531
## alpha[4]	1.62	0.00	0.12	1.39	1.54	1.62	1.70	1.87	2637
## alpha[5]	1.43	0.00	0.13	1.17	1.35	1.43	1.52	1.67	2983
## alpha[6]	1.61	0.00	0.12	1.38	1.52	1.60	1.69	1.86	2834
## alpha[7]	1.50	0.00	0.12	1.26	1.42	1.50	1.58	1.74	3139
## alpha[8]	1.57	0.00	0.11	1.35	1.49	1.57	1.64	1.79	2524
## alpha[9]	1.52	0.00	0.12	1.28	1.44	1.51	1.59	1.75	3465
## alpha[10]	1.57	0.00	0.12	1.34	1.49	1.57	1.65	1.82	3428

```

## alpha[11]    1.57    0.00 0.13    1.33    1.49    1.57    1.66    1.83    3593
## alpha[12]    1.58    0.00 0.12    1.35    1.50    1.58    1.66    1.83    3699
## alpha[13]    1.46    0.00 0.11    1.23    1.38    1.46    1.53    1.68    3140
## alpha[14]    1.43    0.00 0.11    1.20    1.35    1.43    1.51    1.65    3068
## alpha[15]    1.63    0.00 0.13    1.39    1.54    1.62    1.71    1.89    3436
## alpha[16]    1.41    0.00 0.10    1.20    1.34    1.41    1.48    1.60    2833
## alpha[17]    1.38    0.00 0.12    1.14    1.31    1.39    1.46    1.60    3335
## alpha[18]    1.66    0.00 0.10    1.46    1.59    1.66    1.73    1.87    2936
## alpha[19]    1.52    0.00 0.11    1.29    1.45    1.52    1.60    1.75    3122
## alpha[20]    1.62    0.00 0.09    1.43    1.55    1.62    1.68    1.81    2485
## alpha[21]    1.57    0.00 0.09    1.39    1.51    1.57    1.63    1.76    2875
## alpha[22]    1.63    0.00 0.10    1.44    1.56    1.63    1.69    1.82    2594
## alpha[23]    1.51    0.00 0.12    1.27    1.43    1.51    1.60    1.76    3717
## beta1        -0.08    0.00 0.12   -0.31   -0.16   -0.08    0.00    0.16    2514
## beta2        -0.04    0.00 0.04   -0.11   -0.06   -0.04   -0.01    0.03    1925
## beta3         0.13    0.00 0.05    0.03    0.09    0.13    0.16    0.23    3094
## lp__         1892.37    0.12 5.79 1881.29 1888.67 1892.28 1895.85 1904.58 2232
##
## Rhat
## mu           1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1
## alpha[10]    1
## alpha[11]    1
## alpha[12]    1
## alpha[13]    1
## alpha[14]    1
## alpha[15]    1
## alpha[16]    1
## alpha[17]    1
## alpha[18]    1
## alpha[19]    1
## alpha[20]    1
## alpha[21]    1
## alpha[22]    1
## alpha[23]    1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 20:12:17 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 842
UKRN N = 435
Non-UKRN N = 407
Year 1 N = 344
Year 2 N = 273
Placement Year N = 59
Year 4 N = 166

Var1	Freq
0	34
1	53
2	81
3	111
4	150
5	146
6	111
7	77
8	79

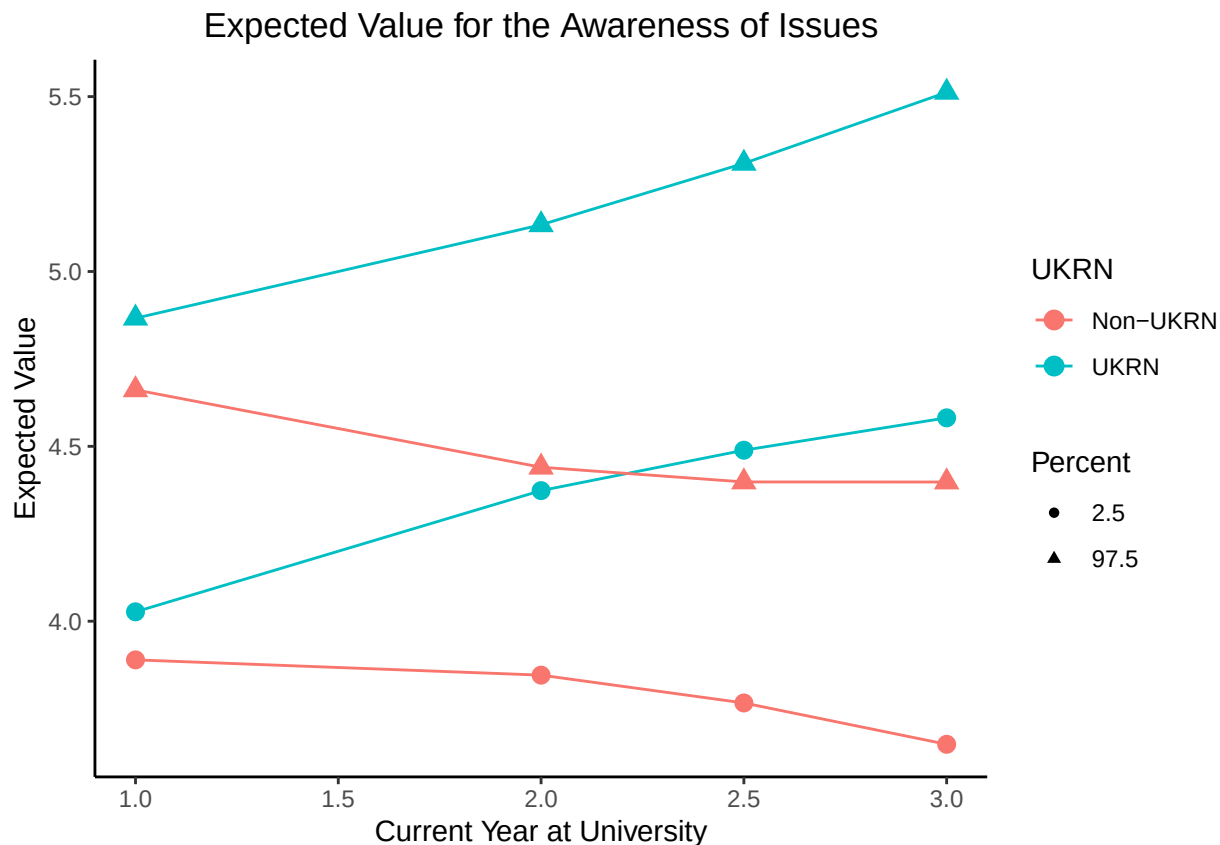
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0267609

97.5% = 0.2286768

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the issues around open and reproducible research in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

117 divergent transitions.



Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.76    0.00  0.10    0.57    0.70    0.76    0.83    0.94   481
## sigma        0.07    0.00  0.04    0.02    0.04    0.06    0.09    0.16   429
## alpha[1]     0.79    0.00  0.11    0.56    0.72    0.79    0.86    0.99   658
## alpha[2]     0.75    0.00  0.12    0.51    0.67    0.75    0.83    0.96   630
## alpha[3]     0.76    0.00  0.11    0.53    0.68    0.76    0.84    0.97   640
## alpha[4]     0.78    0.00  0.12    0.55    0.71    0.78    0.86    1.01   748
## alpha[5]     0.72    0.01  0.12    0.47    0.65    0.73    0.81    0.94   587
## alpha[6]     0.76    0.00  0.12    0.53    0.69    0.77    0.85    0.98   711
## alpha[7]     0.71    0.01  0.13    0.43    0.63    0.71    0.80    0.92   526
## alpha[8]     0.83    0.00  0.12    0.60    0.75    0.83    0.91    1.08   785
## alpha[9]     0.72    0.01  0.12    0.46    0.64    0.73    0.81    0.93   566
## alpha[10]    0.71    0.01  0.13    0.43    0.63    0.72    0.80    0.93   523
```

```

## alpha[11]    0.75    0.00  0.12    0.50    0.67    0.75    0.83    0.97    646
## alpha[12]    0.79    0.00  0.12    0.55    0.71    0.79    0.86    1.02    784
## alpha[13]    0.76    0.00  0.11    0.53    0.68    0.76    0.84    0.97    715
## alpha[14]    0.75    0.00  0.11    0.52    0.68    0.76    0.83    0.96    683
## alpha[15]    0.77    0.00  0.12    0.53    0.70    0.77    0.85    0.99    680
## alpha[16]    0.80    0.00  0.11    0.59    0.73    0.80    0.87    1.01    706
## alpha[17]    0.77    0.00  0.11    0.55    0.70    0.78    0.85    0.98    644
## alpha[18]    0.76    0.00  0.11    0.54    0.69    0.76    0.84    0.95    629
## alpha[19]    0.76    0.00  0.12    0.53    0.69    0.77    0.85    0.98    679
## alpha[20]    0.77    0.00  0.11    0.55    0.70    0.77    0.84    0.96    598
## alpha[21]    0.78    0.00  0.10    0.56    0.71    0.78    0.85    0.97    670
## alpha[22]    0.74    0.00  0.11    0.52    0.66    0.74    0.81    0.93    505
## alpha[23]    0.82    0.00  0.12    0.58    0.74    0.82    0.90    1.08    863
## beta1        -0.14    0.01  0.14   -0.39   -0.23   -0.14   -0.05    0.13    593
## beta2         0.06    0.00  0.04   -0.02    0.03    0.06    0.08    0.14    606
## beta3         0.07    0.00  0.06   -0.06    0.02    0.07    0.11    0.18    787
## lp__         -214.39    0.79 11.13 -233.25 -222.55 -215.59 -206.99 -190.52    198
##              Rhat
## mu            1.01
## sigma          1.01
## alpha[1]       1.00
## alpha[2]       1.01
## alpha[3]       1.01
## alpha[4]       1.00
## alpha[5]       1.01
## alpha[6]       1.01
## alpha[7]       1.01
## alpha[8]       1.00
## alpha[9]       1.01
## alpha[10]      1.01
## alpha[11]      1.01
## alpha[12]      1.00
## alpha[13]      1.01
## alpha[14]      1.01
## alpha[15]      1.00
## alpha[16]      1.00
## alpha[17]      1.01
## alpha[18]      1.01
## alpha[19]      1.01
## alpha[20]      1.01
## alpha[21]      1.00
## alpha[22]      1.01
## alpha[23]      1.00
## beta1          1.01
## beta2          1.01
## beta3          1.00
## lp__           1.02
##
## Samples were drawn using NUTS(diag_e) at Mon Jan  6 20:25:34 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

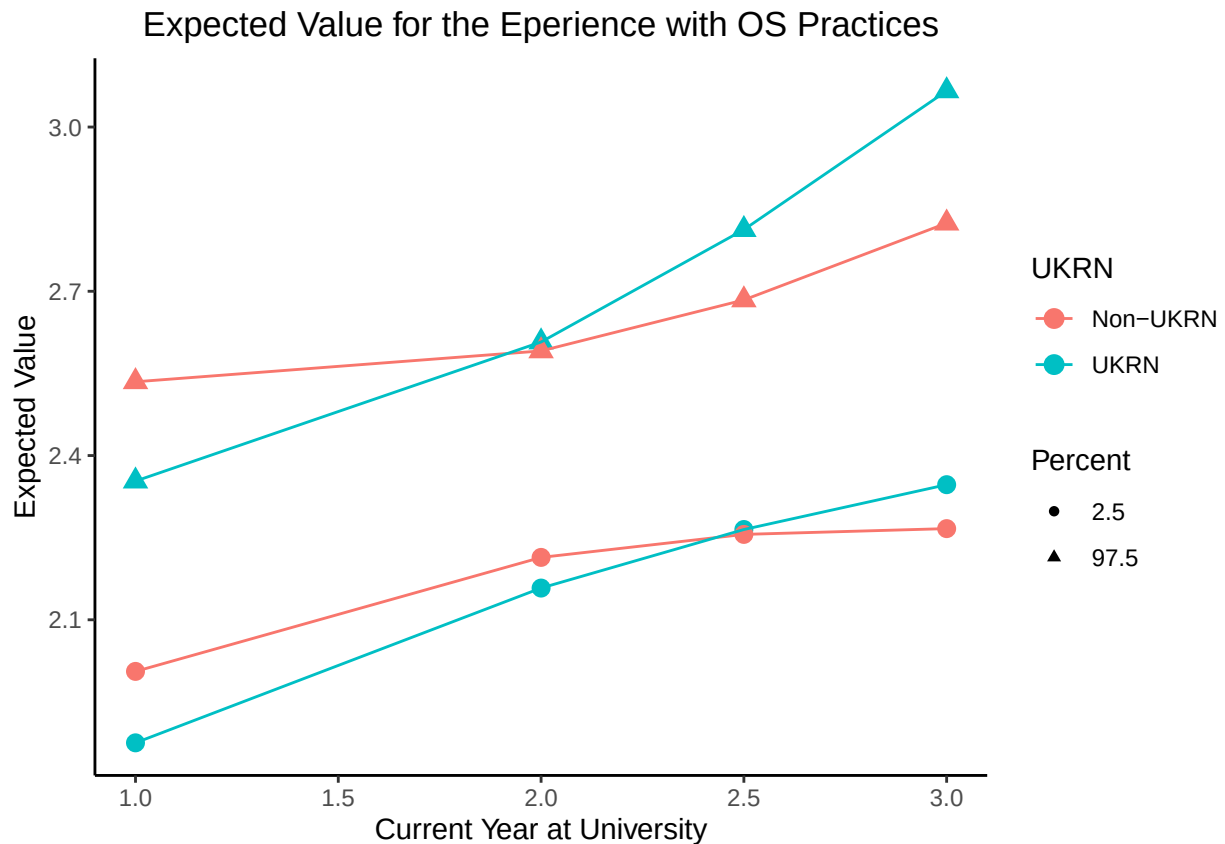
Number of participants in the analysis

Total N = 842
 UKRN N = 435
 Non-UKRN N = 407
 Year 1 N = 344
 Year 2 N = 273
 Placement Year N = 59
 Year 4 N = 166

Var1	Freq
0	138
1	169
2	159
3	165
4	103
5	65
6	28
7	9
8	6

The output shows that all beta credibility intervals cross 0.

1207 divergent transitions. In addition: - There were 2 chains where the estimated Bayesian Fraction of Missing Information was low. - The largest R-hat is 1.02, running the chains for more iterations may help. - Effective Samples Size (ESS) is too low, indicating posterior means and medians may be unreliable. - Effective Samples Size (ESS) is too low, indicating posterior variances and tail quantiles may be unreliable.



Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4. The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu           0.58    0.00 0.05   0.47  0.54  0.58   0.61   0.68  1348 1.00
## sigma        0.35    0.00 0.01   0.33  0.34  0.35   0.35   0.36 11904 1.00
## sigma_a      0.07    0.00 0.03   0.02  0.05  0.07   0.08   0.12   447 1.01
## alpha[1]     0.52    0.00 0.06   0.39  0.48  0.53   0.57   0.65  1701 1.00
## alpha[2]     0.53    0.00 0.08   0.37  0.48  0.53   0.58   0.67  2047 1.00
## alpha[3]     0.57    0.00 0.07   0.44  0.53  0.57   0.62   0.71  1970 1.00
## alpha[4]     0.60    0.00 0.07   0.46  0.55  0.59   0.64   0.74  1941 1.00
## alpha[5]     0.50    0.00 0.08   0.33  0.45  0.51   0.56   0.65  2016 1.00
## alpha[6]     0.68    0.00 0.08   0.54  0.62  0.68   0.74   0.86   851 1.00
## alpha[7]     0.58    0.00 0.07   0.43  0.53  0.58   0.63   0.73  2302 1.00
## alpha[8]     0.61    0.00 0.07   0.48  0.57  0.61   0.66   0.75  1481 1.00
## alpha[9]     0.55    0.00 0.07   0.41  0.51  0.56   0.60   0.69  2574 1.00
## alpha[10]    0.56    0.00 0.07   0.41  0.51  0.56   0.61   0.70  2652 1.00
## alpha[11]    0.59    0.00 0.08   0.44  0.54  0.59   0.64   0.75  2487 1.00
## alpha[12]    0.52    0.00 0.08   0.35  0.47  0.53   0.57   0.66  2527 1.00
## alpha[13]    0.60    0.00 0.07   0.47  0.55  0.60   0.64   0.74  2413 1.00
## alpha[14]    0.53    0.00 0.07   0.39  0.49  0.54   0.58   0.66  2459 1.00
## alpha[15]    0.60    0.00 0.08   0.46  0.55  0.60   0.65   0.76  2305 1.00
## alpha[16]    0.59    0.00 0.06   0.48  0.55  0.59   0.63   0.71  1929 1.00
## alpha[17]    0.55    0.00 0.06   0.42  0.51  0.56   0.60   0.68  2655 1.00
## alpha[18]    0.55    0.00 0.06   0.43  0.51  0.55   0.59   0.67  2766 1.00
## alpha[19]    0.57    0.00 0.07   0.43  0.53  0.57   0.62   0.71  2624 1.00
## alpha[20]    0.63    0.00 0.06   0.52  0.59  0.63   0.67   0.75   998 1.00
## alpha[21]    0.60    0.00 0.06   0.49  0.56  0.60   0.64   0.71  1938 1.00
## alpha[22]    0.62    0.00 0.06   0.50  0.58  0.62   0.66   0.74  1352 1.00
## alpha[23]    0.55    0.00 0.07   0.40  0.50  0.55   0.60   0.69  2654 1.00
## beta1       -0.04    0.00 0.08  -0.18 -0.09 -0.04   0.01   0.12  1572 1.00
## beta2        0.01    0.00 0.02  -0.04 -0.01  0.01   0.02   0.05  1212 1.00
## beta3        0.05    0.00 0.03  -0.01  0.03  0.05   0.08   0.12  1583 1.00
## lp__        518.60    0.57 7.81 505.99 513.68 517.68 522.17 541.21   186 1.02
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 20:39:25 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 835

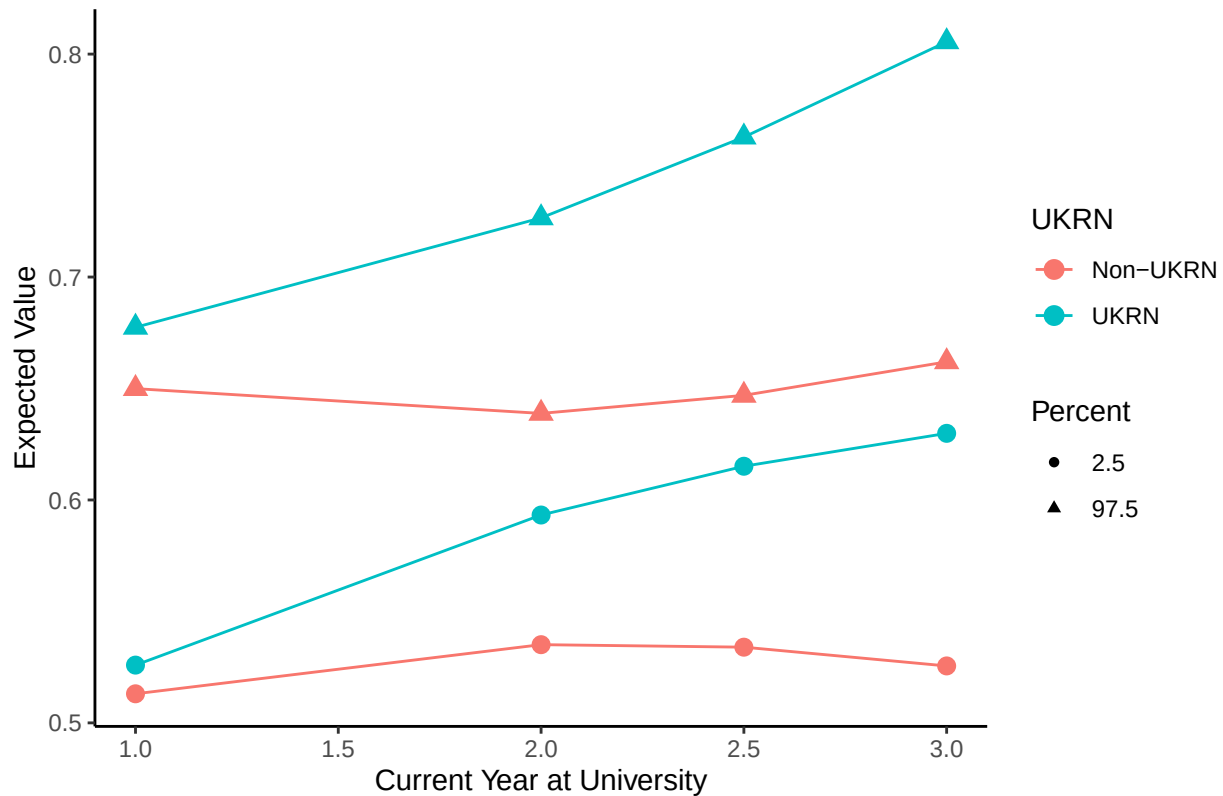
UKRN N = 432

Non-UKRN N = 403
 Year 1 N = 340
 Year 2 N = 270
 Placement Year N = 59
 Year 4 N = 166

The output shows that all beta credibility intervals cross 0.

234 divergent transitions.

Expected Value for positive Conceptual Perceptions of OS Practices



Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4. The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean  sd  2.5%  25%  50%  75%  97.5% n_eff Rhat
## mu          0.36    0.00 0.08   0.20  0.31  0.36  0.42   0.52 2294   1
## sigma       0.48    0.00 0.01   0.46  0.47  0.48  0.49   0.51 18618  1
## sigma_a     0.14    0.00 0.04   0.08  0.12  0.14  0.16   0.23  7628  1
## alpha[1]    0.20    0.00 0.11  -0.01  0.13  0.20  0.27   0.41  2647  1
```

```

## alpha[2]    0.25    0.00 0.12    0.00    0.16    0.25    0.33    0.48 3559    1
## alpha[3]    0.31    0.00 0.12    0.08    0.23    0.31    0.39    0.53 3262    1
## alpha[4]    0.41    0.00 0.12    0.17    0.33    0.41    0.49    0.64 3714    1
## alpha[5]    0.20    0.00 0.13   -0.08    0.11    0.20    0.29    0.44 3689    1
## alpha[6]    0.61    0.00 0.13    0.37    0.52    0.61    0.70    0.89 4445    1
## alpha[7]    0.49    0.00 0.13    0.25    0.41    0.49    0.58    0.75 4426    1
## alpha[8]    0.45    0.00 0.11    0.22    0.37    0.45    0.53    0.68 3392    1
## alpha[9]    0.32    0.00 0.12    0.08    0.24    0.32    0.40    0.55 4953    1
## alpha[10]   0.40    0.00 0.12    0.16    0.32    0.40    0.49    0.65 5135    1
## alpha[11]   0.43    0.00 0.13    0.17    0.34    0.43    0.52    0.69 5218    1
## alpha[12]   0.25    0.00 0.13   -0.01    0.17    0.26    0.34    0.50 5068    1
## alpha[13]   0.40    0.00 0.11    0.19    0.33    0.40    0.48    0.62 5102    1
## alpha[14]   0.31    0.00 0.11    0.08    0.23    0.31    0.38    0.52 4613    1
## alpha[15]   0.44    0.00 0.12    0.20    0.36    0.44    0.52    0.69 5390    1
## alpha[16]   0.37    0.00 0.09    0.19    0.30    0.37    0.43    0.55 4158    1
## alpha[17]   0.25    0.00 0.11    0.04    0.18    0.25    0.32    0.45 4618    1
## alpha[18]   0.37    0.00 0.09    0.19    0.31    0.37    0.43    0.55 4752    1
## alpha[19]   0.28    0.00 0.12    0.05    0.20    0.28    0.36    0.51 4198    1
## alpha[20]   0.44    0.00 0.09    0.26    0.38    0.44    0.50    0.61 3447    1
## alpha[21]   0.40    0.00 0.09    0.22    0.33    0.40    0.46    0.57 4149    1
## alpha[22]   0.51    0.00 0.09    0.34    0.45    0.51    0.57    0.69 3809    1
## alpha[23]   0.29    0.00 0.13    0.03    0.20    0.29    0.38    0.54 5457    1
## beta1       -0.03    0.00 0.12   -0.26   -0.11   -0.03    0.05    0.21 3017    1
## beta2       -0.02    0.00 0.03   -0.08   -0.04   -0.02    0.01    0.05 2519    1
## beta3        0.12    0.00 0.05    0.03    0.09    0.12    0.15    0.22 3945    1
## lp__        218.19    0.06 4.49 208.71 215.36 218.47 221.32 226.10 5587    1
##
## Samples were drawn using NUTS(diag_e) at Mon Jan 6 20:53:25 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 803

UKRN N = 421

Non-UKRN N = 382

Year 1 N = 326

Year 2 N = 261

Placement Year N = 57

Year 4 N = 159

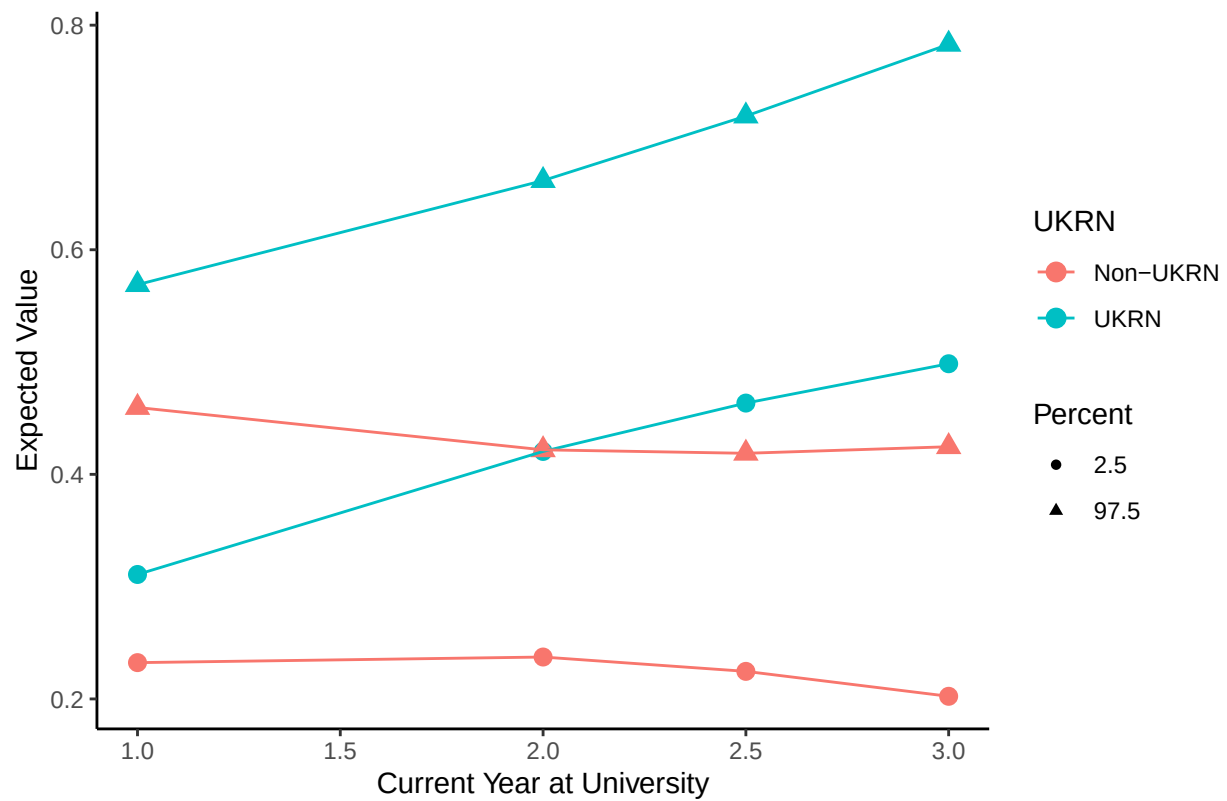
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0270804

97.5% = 0.2150168

The interpretation is that students who attend a UKRN affiliated university are more likely to have positive situational perceptions of open and reproducible practices in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

Expected Value for positive Conceptual Perceptions of OS Practices



Dataset 2 - Beta Bernoulli Model

For the most common category in the Awareness variable

The categories are: * Replication * Open Research Materials * Open Educational Materials * Open Access Journals * Open Data * Power Analysis * Preregistration * Preprint

